

Definition and first terms

$36 A(x) = 35 + 81 x + A(x)^9$ , with the origin-normalized solution.  
 Normalization/grammar:  $A(x)=1+(3)*T(x)$ ;  $T=x+4*T^2+28*T^3+126*T^4+378*T^5+756*T^6+972*T^7+729*T^8+243*T^9$ .  
 $a(0), \dots, a(7) = 1, 3, 12, 180, 3018, 56238, 1121484, 23406804$ .  
 Kernel used throughout:  $\rho(u) = u \left(1 - 4 u^1 - 28 u^2 - 126 u^3 - 378 u^4 - 756 u^5 - 972 u^6 - 729 u^7 - 243 u^8\right)$ , so  $x = \rho(T(x))$ .

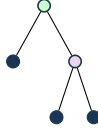
Typogeometry



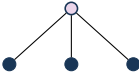
1x3



{1,1}x12



{1,{1,1}}x48



{1,1,1}x84

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities:  $\Delta_2$ : 4,  $\Delta_3$ : 28,  $\Delta_4$ : 126,  $\Delta_5$ : 378,  $\Delta_6$ : 756,  $\Delta_7$ : 972,  $\Delta_8$ : 729,  $\Delta_9$ : 243.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_8) \in \mathbb{Z}_{\geq 0}^8 : \sum_{j=1}^8 j k_j = n - 1\}, \quad K = \sum_{j=1}^8 k_j,$$

$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_8} 4^{k_1} 28^{k_2} 126^{k_3} 378^{k_4} 756^{k_5} 972^{k_6} 729^{k_7} 243^{k_8}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^8 P_r(n) a(n+r) = 0$  for  $n \geq 1$ . The exact degree-8 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 8. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u) / \rho(u)^7$ ,  $\deg_u N = 64$ .  
 Telescoping identity:  $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$ .  
 Matrix dimensions:  $18 \times 18$ ; remainder  $8 \times 9$ , rank 8, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.